Chapter 7: The Rational Numbers.

We will use Equivalence Relations!

Def. 1. Let $\mathbb{Q} = \{(x,y) : x,y \in \mathbb{Z} \ \& \ y \neq 0\}$.

We call x the <u>numerator</u> and y the <u>denominator</u>.

(x,y) is a <u>numerator denominator pair</u>.

Vista: (x,y) will be used to construct $\frac{x}{y}$.

$\frac{1}{2} = \frac{3}{6}$ in $\mathbb{Q}$ $\qquad (1,2) \neq (3,6)$ $\qquad \frac{1}{2} = \frac{3}{6} \Leftrightarrow 1 \cdot 6 = 2 \cdot 3$

$\frac{x}{y} \qquad \frac{z}{w}$

Def. 2. (x,y) and (z,w) in $\mathbb{Q}$ are defined to be <u>rationally equivalent</u> if $xw = yz$. In this case we write $(x,y) \sim (z,w)$.

Thm. 1. $\sim$ is an equivalence relation.

reflexive: $(x,y) \sim (x,y)$, symmetric: $(x,y) \sim (z,w) \Rightarrow (z,w) \sim (x,y)$,

Transitive: If $(x,y) \sim (z,w)$, $(z,w) \sim (r,s)$, then $(x,y) \sim (r,s)$.

Def. 3. We write $[x,y]$ to denote the <u>equivalence class</u> containing (x,y). I.e., $[x,y] = \{(a,b) \in \mathbb{Q} : (a,b) \sim (x,y)\}$

$= \{(a,b) \in Q : ay = bx\}$

Ex.

$$[1,2] = \{(a,b) \in Q : (a,b) \sim (1,2)\}$$
$$= \{(a,b) \in Q : 2 \cdot a = 1 \cdot b\}$$
$$= \{(a, 2a) : a \in \mathbb{Z}, a \neq 0\}$$

Def. 4. Define the <u>rational numbers</u> to be

$$\mathbb{Q} \overset{\text{def}}{=} \{[x,y] : (x,y) \in Q\}$$
$$= \{[x,y] : x, y \in \mathbb{Z}, y \neq 0\}.$$

Q for <u>Quotient</u>

Thm. 2. Let $[x,y] \in \mathbb{Q}$. Then $[x,y] = [0,1]$ if and only if $x = 0$.

Lemma. $[x,y] = [a,b] \iff xb = ay$. $(x,y) \sim (a,b)$

Proof sketch: If $[x,y] = [a,b]$, then $(x,y) \in [a,b]$ since $(x,y) \in [x,y]$. Thus $(x,y) \sim (a,b)$ so $xb = ay$.

If $xb = ay$, then $(x,y) \sim (a,b)$, so $(x,y) \in [a,b]$.

If $(z,w) \in [x,y]$, then $(z,w) \sim (x,y)$. By transitivity $(z,w) \sim (a,b)$, so $(z,w) \in [a,b]$. Thus $[x,y] \subset [a,b]$.

Other direction of containment proceeds similarly.

<u>Proof of Thm. 2.</u> ($\Rightarrow$) If $[x,y] = [0,1]$, then $x \cdot 1 = y \cdot 0$ by Lemma. I.e. $x = 0$. ($\Leftarrow$) Conversely, if $x = 0$, then $x \cdot 1 = 0 = y \cdot 0$. ~~Thus $(x,y) \sim (0,1)$.~~

Thus $[x,y] = [0,1]$ by Lemma. $\square$

<u>Thm. 3.</u> $[x,y] = [1,1] \Leftrightarrow x = y$. $\frac{x}{y} = \frac{1}{1} = 1 \Leftrightarrow x = y$

<u>Def. 5.</u> Define the binary operations $+$ and $\cdot$ on $\mathbb{Q}$ as follows. If $[x,y]$ and $[a,b]$ are in $\mathbb{Q}$

$$[x,y] \cdot [a,b] \overset{\text{def}}{=} [x \cdot a, y \cdot b]$$

and

$$[x,y] + [a,b] \overset{\text{def}}{=} [xb + ay, yb]$$

$$\frac{x}{y} + \frac{a}{b} = \frac{xb}{yb} + \frac{ay}{yb} = \frac{xb+ay}{yb}$$

We gotta prove things are well-defined!

Ex. $[1,2] = [3,6]$ & $[2,3] = [-4,-6]$, so we should have $[1,2]+[2,3] = [3,6]+[-4,-6]$ and

$$[1,2]\cdot[2,3] = [3,6]\cdot[-4,-6]$$

$$\| \qquad\qquad \|$$

$$[2,6] \overset{\checkmark}{=} [-12,-36]$$

$$2\cdot(-36) = 6\cdot(-12)$$

$$[1,2]+[2,3] = [3,6]+[-4,-6]$$

$$\| \qquad\qquad \|$$

$$[3+4,\ 6] = [7,6] \qquad [3\cdot(-6)+(-4)\cdot 6,\ -36]$$

$$\checkmark \qquad\qquad \|$$

$$[-18-24,\ -36]$$

$$\|$$

$$[-42,-36]$$

$$6\cdot(-6)\cdot 7 = 7\cdot(-6)\cdot 6$$

$$6\cdot(-42) = 7\cdot(-36)$$

Adding Wrong

$$\cancel{[1,2]+[2,3]=[3,6]+[-4,-6]}$$

$$\cancel{\begin{array}{ccc} \| & & \| \\ [3,5] & \neq & [-1,6-6]=[-1,0] \end{array}}$$

This is why addition is Hard

Lemmas 4 & 5: $+, \cdot$ are well-defined.

Thm. 6. The additive identity is $[0,1]$

The multiplicative identity is $[1,1]$.

Thm. 7. $[x,y]=[cx,cy]$ if $c\in\mathbb{Z}$ & $c\neq 0$.

Thms 8, 9, 10. $+, \cdot$ are commutative.

and $[x,y]\cdot[y,x]=[1,1]$ if $x,y\neq 0$.

Thm. 11 ($\mathbb{Q}$ is a field) Every element of $\mathbb{Q}$ other than $[0,1]$ has a multiplicative inverse.

<u>Proof.</u> Let $[x,y]\in\mathbb{Q}$ and suppose $[x,y]\neq[0,1]$.

Thus $x\neq 0$. Hence $[y,x]\in\mathbb{Q}$.

We compute $[x,y]\cdot[y,x]=[xy,xy]=[1,1]$.

<u>Def. 6.</u> The <u>canonical embedding</u> $\mathbb{Z}\hookrightarrow\mathbb{Q}$ is defined by

$$a\longmapsto[a,1] \text{ for every } a\in\mathbb{Z}.$$

This allows us to think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$.

Thm. 12. The canonical embedding is injective.

<u>Proof.</u> Suppose $a\mapsto[c,1]$ and $b\mapsto[c,1]$.

Thus $[a,1]=[c,1]$ and $[b,1]=[c,1]$. Hence

$$[a,1]=[b,1], \text{ so } a\cdot 1=b\cdot 1 \text{ or } a=b \checkmark$$

$+,\cdot,-$ all behave well with respect to canonical embedding

$$1+2=3\longmapsto[3,1]$$

Yay!

$$[1,1]+[2,1]=[1\cdot 1+1\cdot 2,1]=[3,1]$$

$$\boxed{[a,1]=a}$$

Subtraction

$$[a,b]-[x,y]\overset{\text{def}}{=}[a,b]+[-x,y]$$

We now think of $\mathbb{Z}$ as living in $\mathbb{Q}$.

If $a\in\mathbb{Z}$ is non-zero, then $\exists\ [1,a]\in\mathbb{Q}$ such that

$$a\cdot[1,a]=[a,1]\cdot[1,a]=[a,a]=[1,1]=1.$$

We see that every non-zero integer has a multiplicative inverse in $\mathbb{Q}$. We denote the multiplicative inverse of a by a^{-1} or $\frac{1}{a}$. We see that any

$$a^{-1}=\frac{1}{a}=[1,a]$$

rational number $[x,y]$ can be written as $[x,1]\cdot[1,y]$ $=x\cdot y^{-1}=x\cdot\frac{1}{y}$. Thus we introduce the notation $[x,y]=\frac{x}{y}$.

Thus $\mathbb{Q} = \{ \overset{[a,b]}{\overset{\shortparallel}{\frac{a}{b}}} : a, b \in \mathbb{Z} \ \& \ b \neq 0 \}$

Thm. 20 + Lemma 21 + Thm. 22. Let $r \in \mathbb{Q}$, then ~~either $r = 0 = [0,1]$, or~~ there are unique integers $a, b \in \mathbb{Z}$ such that $r = \frac{a}{b}$ with $b > 0$ and $\gcd(a,b) = 1$.

<u>Proof</u>: Let $r \in \mathbb{Q}$. Thus $r = [x,y]$ with $y \neq 0$. Let $d = \gcd(x,y)$. We have $x = d \cdot a$ and $y = d \cdot b$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$ and $\gcd(a,b) = 1$.

If $\gcd(a,b) = f > 1$, then $x = d \cdot f \cdot a_0$ & $y = d \cdot f \cdot b_0$, so with $a_0, b_0 \in \mathbb{Z}$ $\Rightarrow$ ✗ df is a common divisor of x, y that is greater than d.

We see that $r = [x,y] \overset{\text{def}}{=} [a,b] = \frac{a}{b}$ since $xb = dab = ay$.

If $b < 0$, then take $[-a, -b]$ so, renaming if necessary, we assume $b > 0$. Thus

$$\mathbb{Q} = \{ \tfrac{a}{b} : a, b \in \mathbb{Z}, b > 0, \gcd(a,b) = 1 \}$$

To see this representation is unique suppose

$\frac{a}{b} = \frac{c}{d}$ with $a,b,c,d \in \mathbb{Z}$, $\gcd(a,b) = \gcd(c,d) = 1$ and $b,d > 0$.

Hence $ad = bc$. We claim that since $\gcd(a,b)=1$ and $a \mid bc$, $a \mid c$. For the same reason $d \mid b$.

Euclid's Lemma. If $\gcd(r,s)=1$ and $r \mid st$, then $r \mid t$. Proof. By Bézout's $\exists\, x,y \in \mathbb{Z}$ such that $rx + sy = 1$. Multiplying by t, we have $trx + tsy = t$. Since $r \mid st$, we see $r \mid (rtx + sty)$. Hence $r \mid t$. $\square$

$ad = bc$. By Euclid's lemma $a \mid c$ & $d \mid b$. That is, $\exists\, k,m \in \mathbb{Z}$ such that $c = ak$ & $b = dm$.

Hence $ad = bc$ becomes $ad = akdm$.

If $ad = 0$, then $a = 0$ since $d > 0$

So since $b > 0$ and $ad = bc$, we have $c = 0$.

Since $\gcd(a,b) = \gcd(0,b) = b$ and $\gcd(c,d) = \gcd(0,d) = d$ we see $b = d = 1$. Thus $a = c = 0$ and $b = d = 1$.

Assume now that $a, c \neq 0$. Thus $ad \neq 0$ since $d > 0$.

Using cancellation $ad = akdm$ becomes

$ad \cdot 1 = ad \cdot km$

$$1 = k \cdot m.$$

Thus $k = m = 1$ or $k = m = -1$

Since $b = dm$. and $b, d > 0$, we see $m \neq -1$, so $k = m = 1$.

Therefore $c = a \cdot 1$ & $b = d \cdot 1$

and our representation $r = \frac{a}{b}$ with $\gcd(a,b) = 1$, $b > 0$ is unique!